

Optimized S-Curve Motion Profiles for Minimum Residual Vibration

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Abstract

A method for developing optimized point-to-point motion profiles to achieve fast motions with minimum vibration is presented. The proposed approach uses the well-known s-curve motion profiles, but optimizes the selection of the ramp-up (and ramp-down) time. The selection of ramp-up time is based on a frequency analysis that minimizes the excitation energy of the input forcing function at the system natural frequency. Simulation results on a lightly-damped system undergoing point-to-point motions demonstrate that the proposed approach decreases residual vibration by almost an order of magnitude over other approaches, even when the actual natural frequency is in error by 10%.

1. Introduction

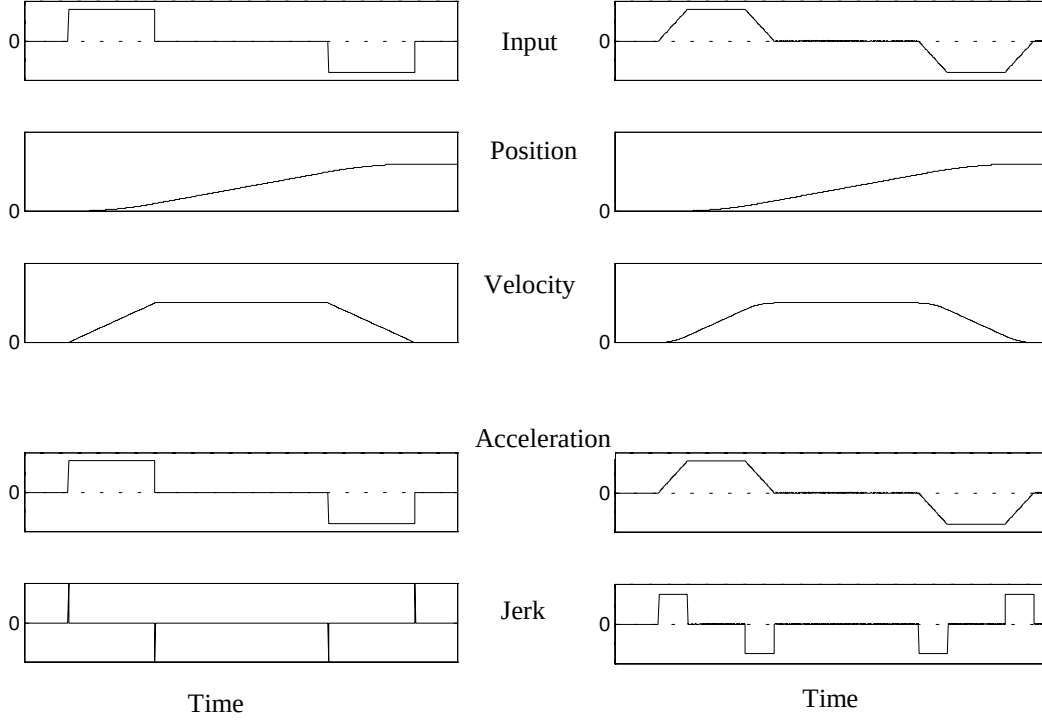
The challenge of achieving fast motions without residual vibration pervades many modern manufacturing applications. It is well known that simple trapezoidal velocity profiles, in which the machine is accelerated to a constant velocity at a constant acceleration and decelerated to rest at a constant deceleration, can achieve fast motions. However, since most real systems exhibit some compliance, these reference profiles tend to excite vibrations that require time to dissipate for the final position to be reached with desired precision. As a result, s-curve velocity profiles have been proposed to reduce the tendency to excite system vibration [1,2]. In these profiles, the trajectory ramps up to peak acceleration and ramps down to constant velocity, thereby producing an s-shaped velocity profile. Depending on the duration of this ramp-up time, the resulting motion can be both fast and precise. However, to date, no systematic method for optimally tuning this parameter has been presented. This paper will extend earlier work [3,4,5] on shaping the frequency content of the input commands to optimally select the ramp-up time for s-curve velocity profiles. Both rest-to-velocity and rest-to-rest simulation results for a lightly-damped system will be provided.

Typical motion profiles for trapezoidal and s-curve velocities are shown in Fig. 1. The input is typically a force or torque command. Notice that the trapezoidal velocity profiles exhibit large spikes in jerk, whereas the s-curve profiles exhibit finite jerk spread out over a period of time. It is primarily this quality that contributes to lower vibration for the s-curve profiles. However, it can be shown that even trapezoidal profiles can achieve vibration-free motion for highly flexible systems if the acceleration and deceleration periods are properly chosen. Thus, the ultimate vibration performance depends on more than just the presence or absence of large spikes in jerk. In addition, without further analysis, it is impossible to determine an optimal selection of the ramp-up time (and ramp-down time) for the s-curve profiles that will achieve both fast motions and minimum residual vibration.

2. Frequency Analysis

An undamped two-mass system, with natural frequency $\omega_n = \sqrt{(1 + m_1/m_2)k/m_1}$, shown in Fig. 2, is used to develop desired motion profiles. Although real systems exhibit some dissipation, the results presented here apply to lightly-damped systems with damping ratio less than 0.1. Most real systems have inherent damping ratios on this order. For this analysis, only the acceleration to peak velocity will be investigated. By properly constructing the acceleration profile, the system will reach peak velocity with minimal vibration. It will then coast at peak velocity. The same waveform with negative force can be utilized to decelerate the system to rest, thereby achieving a similar suppression of residual vibration once the system comes to rest.

The forcing function inputs needed to generate the acceleration portion of the trapezoidal and s-curve velocity profiles are shown in Fig. 3, along with their corresponding parameters. By symmetry, the ramp-down time to reach



(a) Trapezoidal Input

(b) S-Curve Input

Fig. 1 Comparison of Trapezoidal and S-Curve Velocity Profiles

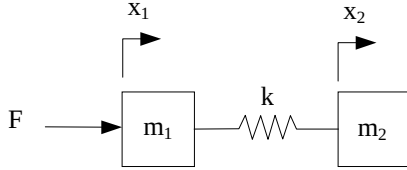


Fig. 2 System Model

constant velocity has the same duration as t_a , the ramp-up time. Note also that when $t_a = 0$, the input for an s-curve profile matches the rectangular pulse input for the trapezoidal profile. In fact, the total time t_{sp} to achieve peak velocity with the s-curve profile can be related to the total time t_r for the rectangular pulse as follows:

$$t_{sp} = \frac{t_r}{1 - t_a^*} \quad (1)$$

where $t_a^* = t_a / t_{sp}$. The rectangular pulse time t_r is related to the peak force F_{max} , total mass m_{total} , and peak velocity v_p as follows:

$$t_r = \frac{m_{total} v_p}{F_{max}} \quad (2)$$

The acceleration amplitude when the peak velocity is reached and the input is turned off is used as a measure of residual vibration. This vibration amplitude can be related to the frequency content of the forcing function input that is used to accelerate the system [5]. For the s-curve force input, this relationship is given by:

$$a^* = \left| \frac{e^{-j\omega_n t_r t_a^* / (1 - t_a^*)} + e^{-j\omega_n t_r} - e^{-j\omega_n t_r / (1 - t_a^*)} - 1}{\omega_n t_r t_a^* / (1 - t_a^*)} \right| \quad (3)$$

where the dimensionless acceleration amplitude $a^* = a m_{total} / F_{max}$, m_{total} is the total inertia being moved and F_{max} is the peak force used for acceleration. As this expression makes clear, the residual vibration amplitude is a strong function of the dimensionless ramp-up time t_a^* . The parameter $\omega_n t_r / 2\pi$, hereafter referred to as the dimensionless response time, represents the total number of periods of oscillation that occur within the rectangular pulse duration t_r . As such, it reflects the combined effects of system flexibility (ω_n) relative to the speed of response t_r . Systems having more compliance and faster move times will tend to have smaller values of $\omega_n t_r / 2\pi$.

Fig. 4 shows dimensionless residual acceleration amplitudes a^* (expressed in decibels) for the undamped two-mass



Fig. 3(a) Force input to generate acceleration portion of trapezoidal velocity profile

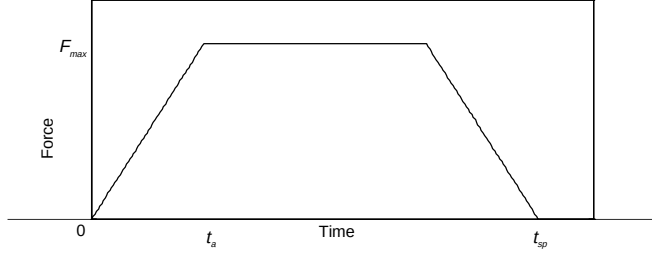


Fig. 3(b) Force input to generate acceleration portion of s-curve velocity profile

system as function of t_a^* for several values of $\omega_n t_r / 2\pi$. Of particular interest are the dips in acceleration amplitude that occur at several values of t_a^* . The dip corresponding to the smallest value of t_a^* minimizes residual acceleration while providing the fastest response relative to the dips having higher values of t_a^* . Notice that the value of t_a^* which minimizes response time becomes smaller as the value of $\omega_n t_r / 2\pi$ increases. Also, the number of dips in the acceleration amplitude plot increases for increasing values of $\omega_n t_r / 2\pi$.

As a careful look at the relationship in eq. (3) shows, the acceleration amplitude is identically zero when $\omega_n t_r / 2\pi$ takes on integer values and $t_a^* = 0$. This implies that a conventional rectangular force input suffices for integer values of $\omega_n t_r / 2\pi$. However, when the actual value of $\omega_n t_r / 2\pi$ changes even slightly away from an integer value (as is shown with $\omega_n t_r / 2\pi = 2.01$ in the second plot of Fig. 4), a value of $t_a^* = 0$ generates significant acceleration. Thus, for robustness to uncertainty in natural frequency, the value of t_a^* chosen for integer values of $\omega_n t_r / 2\pi$ corresponds to the next dip in the acceleration amplitude curve (a t_a^* value of 0.332 for $\omega_n t_r / 2\pi = 2$).

3. Development of Optimized S-Curve Profiles

If the minimum value of t_a^* that generates a dip in residual vibration (as shown in Fig. 4) is found for each value of $\omega_n t_r / 2\pi$, the plot of Fig. 5 results. Note that smaller values of t_a^* are suggested for higher values of $\omega_n t_r / 2\pi$.

The values of t_a^* given in Fig. 5 will reduce both response time and residual vibration. For this reason, they have been called “optimal.” In practice, many control engineers have

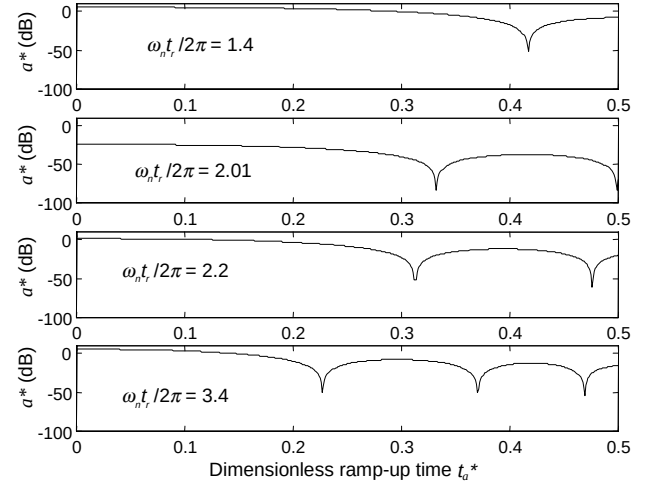


Fig. 4 Variation of a^* with t_a^* for various $\omega_n t_r / 2\pi$ values

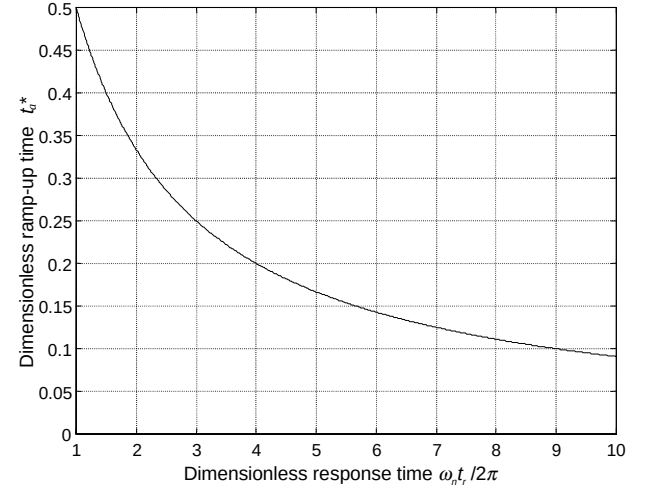


Fig. 5 Optimal t_a^* values as function of $\omega_n t_r / 2\pi$

used a “Rule of Thumb” that suggests picking $t_a^* = 1/6$ [1]. To compare these cases, residual acceleration amplitudes have been computed for various values of the parameter $\omega_n t_r / 2\pi$. The results are given in Fig. 6 (with residual acceleration amplitude again expressed in decibels). Three different forcing function inputs are compared for the undamped system of Fig. 2; using rectangular pulse inputs with $t_a^* = 0$ that generate trapezoidal velocity profiles, using s-curve profiles with $t_a^* = 1/6$, and using s-curve profiles with optimal t_a^* values chosen as a function of $\omega_n t_r / 2\pi$ based on Fig. 5. Notice that even the simple s-curve input using $t_a^* = 1/6$ improves upon the rectangular pulse input for most cases. The “optimal” selection of t_a^* gives the lowest residual vibration over the entire range of $\omega_n t_r / 2\pi$ values, with significant improvement over the other approaches. The “noisiness” of the data is a result of numerical precision in correctly determining the true minimizing value of t_a^* .

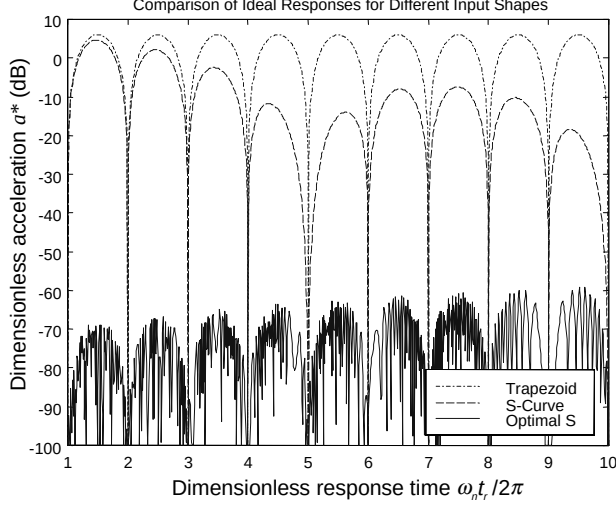


Fig. 6 Comparison of dimensionless residual vibration amplitude for different inputs for an ideal undamped system having a single vibrating mode

The results in Fig. 6 look promising, but these assume that the system parameters, in particular the natural frequency, are known precisely. Since the method for picking ramp-up time t_a^* relies on finding the true minimum residual acceleration, the results of Fig. 4 suggest that this minimum is very narrow. Thus, a small change in system parameter may well cause large residual acceleration. To check this, the comparison plot of Fig. 6 was rerun, again for the undamped system, but this time, the actual natural frequency was assumed to be 10% higher than the natural frequency for which the optimized s-curve profiles were created. The results for the same three input functions (with $t_a^* = 0$, $t_a^* = 1/6$, and optimal t_a^*) are shown in Fig. 7. Although the residual accelerations for the optimized s-curve inputs are larger than for the ideal case in Fig. 6, they are still significantly lower than for either the trapezoidal input or the s-curve input with $t_a^* = 1/6$. In fact, the most significant improvement results at low values of $\omega_n t_r / 2\pi$, which represent systems that have a low natural frequency or require extremely fast motions, a regime that is usually the most difficult for achieving vibration-free motions. The optimized s-curve profiles represent a viable strategy for achieving these fast motions.

4. Point-to-Point Simulations

Since the focus of this investigation is to achieve vibration-free point-to-point motions, the previous optimized s-curve acceleration profiles must be incorporated into a total motion profile that accelerates, coasts, and decelerates, as shown in Fig. 1. Since the same forces as were used for acceleration are used to decelerate the system, the total time spent in decelerating from the peak velocity is the same as

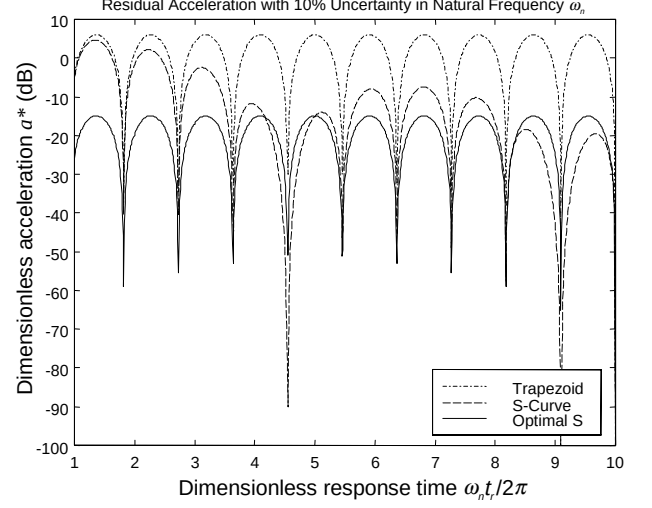


Fig. 7 Comparison of dimensionless residual vibration amplitude for different inputs for an undamped system with an actual natural frequency 10% higher than nominal.

that spent to accelerate. The only additional parameter is the duration of time spent coasting at peak velocity. In keeping with previous practice, this time is nondimensionalized as $\omega_n t_d / 2\pi$, that is, it represents the number of oscillation periods during the time t_d . Obviously, the total duration of the input forcing function will depend on the desired point-to-point displacement y_f . The duration of coast time is related to this displacement as follows:

$$t_d = \frac{y_f}{v_p} - t_r = \frac{y_f}{v_p} - \frac{m_{total} v_p}{F_{max}} \quad (4)$$

Note that the relationship between final displacement y_f and peak velocity v_p determines whether there will be any constant velocity coasting phase. For short displacements and high peak velocities, the above calculation for t_d would become negative, thus suggesting that the velocity profile is triangular, that is, the system must decelerate even before it reaches peak velocity. In that case, the peak velocity must be reduced until t_d just approaches zero, and that reduced value of peak velocity must then be used to determine t_r and the corresponding optimized force profile.

A series of simulations have been performed to evaluate the residual acceleration amplitudes for the same three input functions as before, but this time generating a total point-to-point motion. The same two-mass system as shown in Fig. 2 was used but this time a small amount of damping (damping ratio ζ of 0.05) was added. The transfer function relating x_2 to F is given by:

$$\frac{X_2(s)}{F(s)} = \frac{2\zeta \left(\frac{\omega_a}{\omega_n} \right) s + \left(\frac{\omega_a}{\omega_n} \right)^2}{s^2 \left[s^2 + 2\zeta \left(\frac{\omega_a}{\omega_n} \right) s + \left(\frac{\omega_a}{\omega_n} \right)^2 \right]} \quad (5)$$

where ω_a is the actual natural frequency, ω_n is the nominal natural frequency, and s is normalized by ω_n . The simulations capture the effect of damping as well as the effect of the vibration remaining at the end of the coast phase on the residual vibration at the end of the move, when the final nominal position has been reached. A coast time of $\omega_n t_d / 2\pi = 10$ was used for all simulations.

Figure 8 shows the simulated peak residual acceleration amplitudes for the ideal system (when $\omega_a = \omega_n$) for the three different input profiles (with $t_a^* = 0$, $t_a^* = 1/6$, and optimal t_a^*). Notice that even the trapezoidal input generates acceleration amplitudes that diminish with higher values of $\omega_n t_r / 2\pi$. This is understandable, since higher values of $\omega_n t_r / 2\pi$ imply that more cycles of oscillation take place within the duration of the motion. Thus, more periods of natural dissipation from damping are available to diminish the residual acceleration amplitudes. The simulations once again demonstrate the value of using the optimized s-curve profiles to generate motion. These inputs consistently result in lower values of residual acceleration than the other two input profiles.

Whereas the results of Fig. 8 were generated using an ideal system model, another series of simulations were performed with the simulated natural frequency a full 10% higher than the nominal natural frequency for which the optimized input profiles were generated (i.e., $\omega_a = 1.1 \omega_n$). These results are shown in Fig. 9, again for all three input profiles. Although the residual acceleration amplitudes for the optimized inputs are higher than for the ideal case of Fig. 8, they are still significantly lower than the amplitudes for the other two inputs for a major portion of the range of $\omega_n t_r / 2\pi$ values. The improvement is on the order of 20 dB at small values of $\omega_n t_r / 2\pi$, where system natural frequency is low and response time is fast.

5. Conclusions

Motion control using the so-called s-curve velocity profile has become a popular approach to achieving fast motions with reduced residual vibration. However, the ramp-up time to reach peak acceleration for these forcing profiles has never been optimized to take full advantage of the vibration-reducing capability of this forcing function. This paper provides a method based on the frequency content of the forcing function to select this ramp-up time so that both response time and residual vibration amplitude are minimized. Open-loop simulations for a lightly-damped system confirm that these optimized input profiles can achieve superior performance, even when the actual system natural frequency differs by 10 % from the frequency for which the inputs were computed. Vibration reduction of almost an order of magnitude is possible, relative to non-optimized inputs. This improvement is most pronounced for systems having low natural frequencies and requiring fast motions.

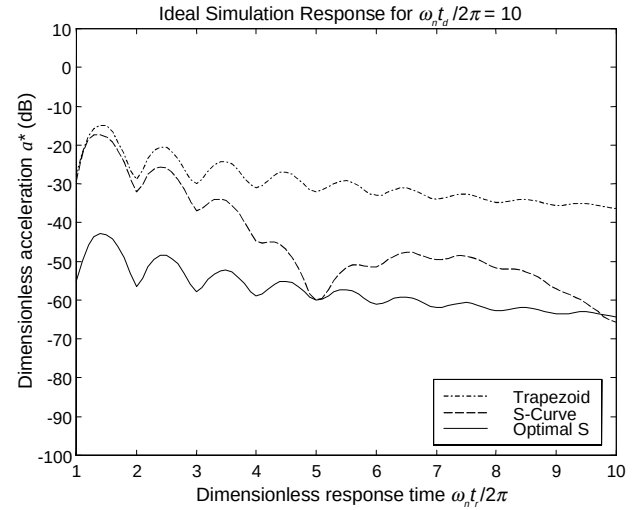


Fig. 8 Simulation results showing peak residual vibration amplitude for a lightly-damped system with ideal natural frequency using three different input profiles.

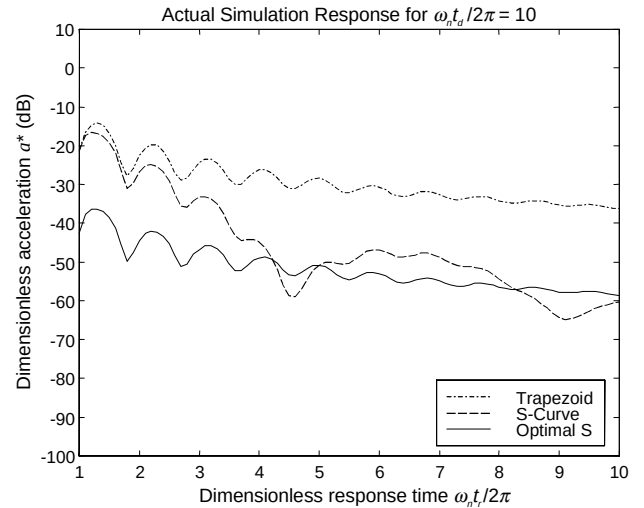


Fig. 9 Simulation results showing peak residual vibration amplitude for a lightly-damped system with natural frequency that is 10% higher than nominal.

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